

Universal Employee Verification & Multi-Company Attendance System

A unified, dual-purpose enterprise solution combining cryptographic ID barcode identity passes, real-time optical kiosk attendance, permission lifecycle tracking, and complete month-wise lifetime company attendance analytics.

Author: Engineering Team • **Database:** PostgreSQL (Prisma ORM) •

Architecture: Single-Page Application (SPA) + Express API • **Release Date:** August 2026

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🔑 1. Executive Overview & Core Problem Statement

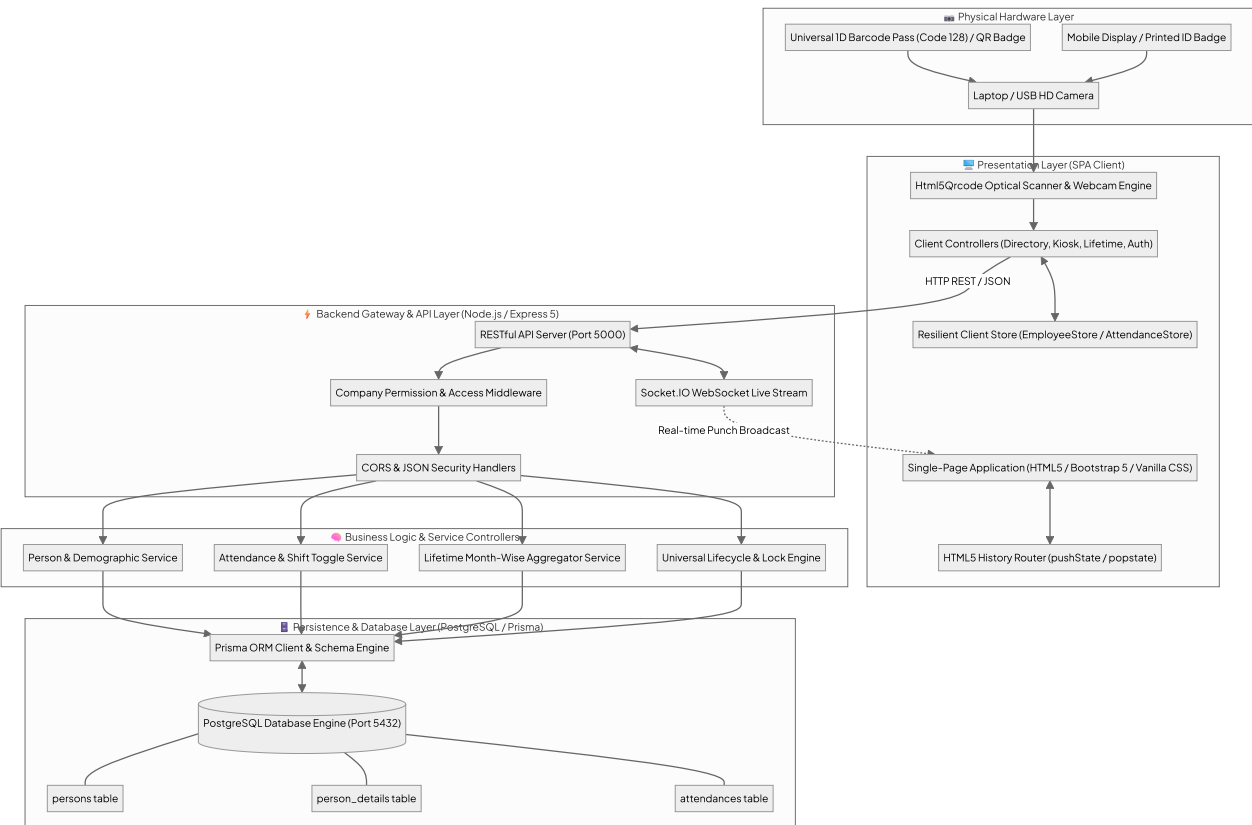
Traditional workforce management systems suffer from disconnected silos: employee background verification data lives in static reports, while daily attendance is locked inside standalone biometric hardware. When an employee transitions between companies, their verified identity is lost, and their historical attendance reliability cannot be verified.

The Universal Employee Verification & Attendance System solves this by establishing a Universal Barcode Identity:

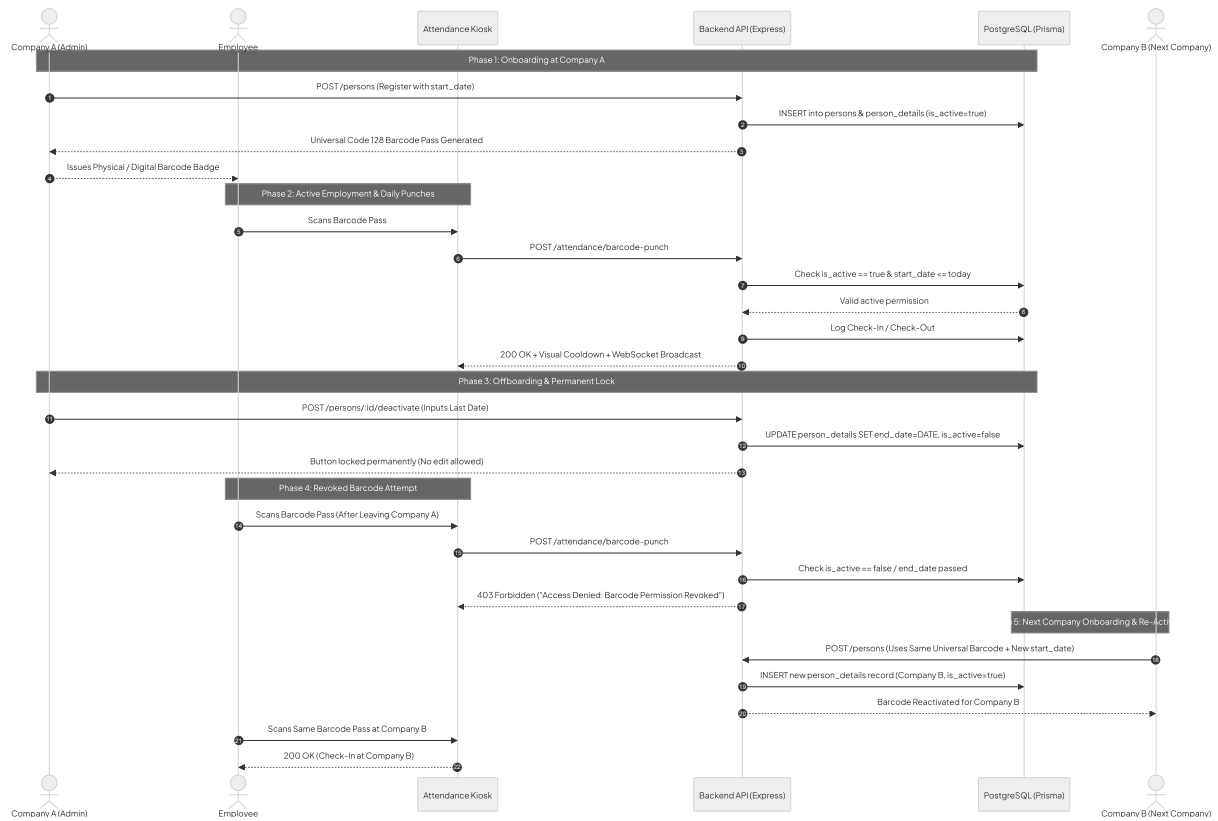
- **Permanent Cryptographic Barcode Pass:** Generated once per employee, encoding their unique system identifier.
- **Company-Level Permission Management:** A company grants active access when onboarding and revokes access upon entering a permanent, unmodifiable *Last Date*.
- **Multi-Tenure Re-Activation:** When an employee moves to their next company, the new company activates the *exact same universal barcode pass*, creating a unified cross-company career record.
- **Lifetime Month-Wise Attendance Audits:** Companies can scan the barcode pass to audit verified monthly counts for **Total Present**, **Total Absent**, and **Permissions & Leaves** across the entire tenure.

❁ 2. High-Level System Architecture & Flow

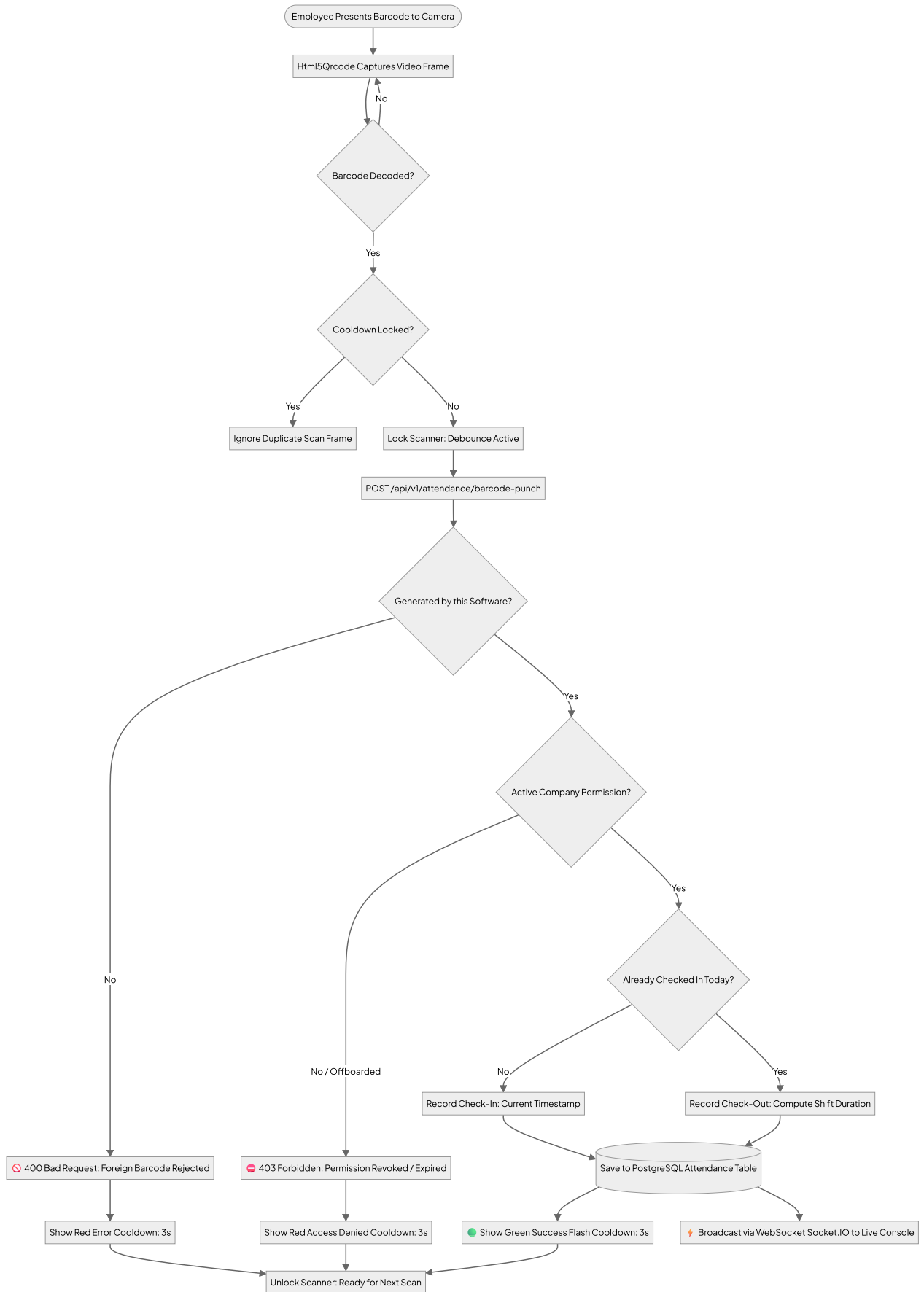
🔗 Full-Stack Enterprise Tiered Architecture



Universal Barcode Permission Lifecycle Flow



Optical Kiosk Scan Decision & Cooldown Engine



End-to-End Information & Event Flow

- 1 Identity Generation & Verification Store:** Admin registers employee demographics, salary history, and salary slip PDF. System creates records in PostgreSQL (**Person** & **PersonDetails**) and renders a Code 128 / QR digital pass.
- 2 Company Attendance Kiosk (Daily Clock-In/Out):** Employee presents their universal barcode to the camera kiosk. The optical scanner decodes the barcode, validates company permission, logs Check-In / Check-Out, triggers a 3-second visual cooldown, and broadcasts live scans via WebSocket to the live activity stream.
- 3 Lifecycle Offboarding & Permanent Lock:** When an employee departs, the company enters their *Last Date*. Once submitted, the field is permanently locked, and the barcode is switched to inactive (403 Forbidden on attendance punch).
- 4 Next Company Onboarding:** The subsequent company looks up the existing barcode pass, inputs their new joining record (start date, role, department), and reactivates the same universal barcode for their company facility.
- 5 Lifetime Attendance Analytics Audit:** Any authorized company scans or searches the barcode to query PostgreSQL, grouping attendance punch history by month and displaying verified monthly Present, Absent, and Permission rates.

3. Module 01: Universal Barcode Identity & Generation

The generation engine converts verified demographics into a scannable Code 128 standard 1D barcode and 2D QR badge.

Attribute	Specification	Purpose
Symbology	Code 128 (High-Density Alphanumeric) & QR Code	Ensures instant readability on all 1D laser scanners and webcams.
Universal Payload	EMP-XXX or Cryptographic Hash	Resolves to the employee's immutable primary key in the database.
Badge Components	Photo, Universal Code, Active Company, Role, Salary Slip URL	Printable digital credential for on-site visual and hardware verification.

4. Module 02: Real-Time Optical Attendance Kiosk

The kiosk module turns any laptop, tablet, or USB camera into an enterprise-grade biometric barcode terminal.

Key Kiosk Capabilities:

- **Smart Toggle Check-In / Check-Out:** First scan of the day logs *Check-In*; subsequent scans log *Check-Out* and compute exact shift duration.
- **Foreign Barcode Rejection:** Only barcodes generated by this software are validated; random retail barcodes are rejected immediately.
- **3-Second Visual Cooldown Overlay:** Prevents duplicate scanning loops and displays employee name, photo, and time badge.
- **Real-Time WebSocket Stream:** Live scans appear instantly in the manager activity console without page reloads.

5. Module 03: Barcode Permission Lifecycle (Locking & Re-Activation)

Addresses strict enterprise governance for employee joining, offboarding, and career transitions:

Lifecycle Stage	System Action	Database & Permission State
1. New Onboarding	Admin assigns barcode + Start Date	<code>is_active = true, start_date = DATE, end_date = NULL.</code> Active for attendance.
2. Offboarding	Admin clicks <i>"Enter Last Date"</i>	<code>end_date = DATE, is_active = false.</code> Button permanently replaced with locked badge ( Last: YYYY-MM-DD).
3. Access Revoked	Offboarded employee scans barcode	Terminal returns 403 Access Denied ("Barcode permission revoked or expired").
4. Next Company	New company activates barcode	Creates new <code>PersonDetails</code> entry with new joining date and company facility, reactivating pass.



6. Module 04: Lifetime Company Attendance (Month-Wise Breakdown)

Allows companies to audit an employee's lifetime performance at their facility with month-by-month breakdowns:

Month-Wise Audit Parameters:

Total Present

Verified physical kiosk check-ins and remote punch records for the month.

Total Absent

Unexcused absences and recorded leave days during the working period.

Permissions & Leaves

Late arrivals, approved permission slips, and half-day authorizations.

Reliability Rate

Calculated attendance percentage: $(\text{Present} / \text{Total Working Days}) * 100$.

7. Relational Database Schema & Data Models

The database is built on PostgreSQL via Prisma ORM to maintain strict relational integrity across employment history and attendance logs:

```
model Person {
  id String @id @default(uuid())
  name String
  email String @unique
  mobileNumber String?
  @map("mobile_number")
  employeeCode String @unique
  @map("employee_code")
  address String?
  photoUrl String?
  @map("photo_url")
  status String @default("active")
  createdAt DateTime @default(now())
  @map("created_at")
  details PersonDetails[]
  attendances Attendance[]
  @@map("persons")
}
model PersonDetails {
  id String @id @default(uuid())
  personId String @map("person_id")
  companyName String @map("company_name")
  department String?
  roleName String @map("role_name")
  startDate DateTime @map("start_date")
  endDate DateTime? @map("end_date")
  isActive Boolean @default(true)
  @map("is_active")
  monthlySalary Decimal
  @map("monthly_salary")
  annualSalary Decimal
  @map("annual_salary")
  barcodeData String?
  @map("barcode_data")
  salarySlipUrl String?
  @map("salary_slip_url")
  salarySlipName String?
  @map("salary_slip_name")
  person Person @relation(fields: [personId], references: [id], onDelete: Cascade)
  @@map("person_details")
}
model Attendance {
  id String @id @default(uuid())
  personId String? @map("person_id")
  employeeCode String @map("employee_code")
  employeeName String @map("employee_name")
  department String?
  punchDate DateTime @map("punch_date")
  @db.Date
  punchTime DateTime @default(now())
  @map("punch_time")
  checkInTime DateTime? @map("check_in_time")
  checkOutTime DateTime?
  @map("check_out_time")
  status String @default("Present")
  workMode String? @default("On-Site Kiosk")
  @map("work_mode")
  verificationType String? @default("Barcode Scanner")
  @map("verification_type")
  notes String?
  person Person? @relation(fields: [personId], references: [id], onDelete: SetNull)
  @@map("attendances")
}
```

API 8. REST API Specification & Endpoint Directory

Method	Endpoint	Functionality
POST	/api/v1/persons	Registers new employee, creates demographics & employment history.
GET	/api/v1/persons	Retrieves all registered employees with relational employment histories.
POST	/api/v1/persons/:id/deactivate	Offboards employee by recording unmodifiable <code>end_date</code> and setting <code>is_active=false</code> .
POST	/api/v1/attendance/barcode-punch	Validates barcode pass, checks active company permissions, records check-in/out.
POST	/api/v1/attendance/employee-lifetime-analytics	Aggregates lifetime punch logs into month-wise Present, Absent, and Permission metrics.
GET	/api/v1/attendance/records	Fetches workforce attendance punch history with durations.
DELETE	/api/v1/attendance/records/:id	Removes specific attendance log entry.